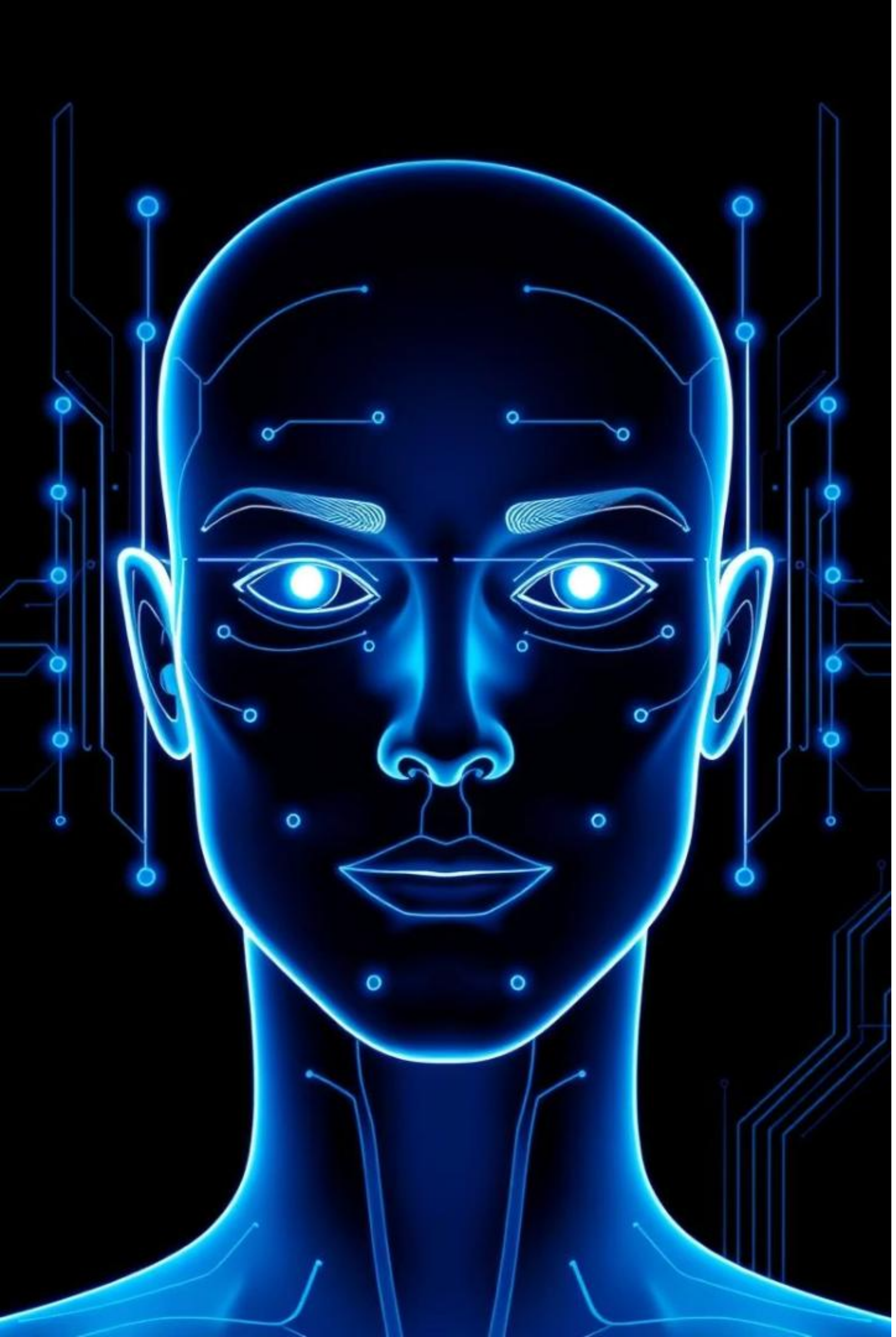

Face Recognition Attendance System

Automating Attendance Tracking with AI

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Problem Statement

Manual System Issues

- Time-consuming roll calls delay classes
- Proxy attendance by friends
- Human errors and lost records

Our AI Solution

- AI-based facial recognition
- Zero physical contact, post-pandemic safe

Key Objectives

Automate Attendance

Real-time face detection replaces paper

High Accuracy

Over 85% recognition rate with LBPH

Prevent Fraud

Biometric verification stops proxy attendance

User-Friendly Interface

Simple GUI to add students and view logs

Scalability

Supports 1000+ students with MySQL

System Workflow

1

Admin Adds Student

Uploads photo, roll number, department

2

Face Data Collection

Captures 100 images per student

3

Model Training

LBPH creates unique face signature

4

Real-Time Detection

Camera matches faces with database

5

Attendance Logged

Auto-saves to CSV/MySQL with timestamp

Technical Stack

Face Detection

OpenCV Haar Cascade

Face Recognition

LBPH Algorithm: lightweight, accurate

Database

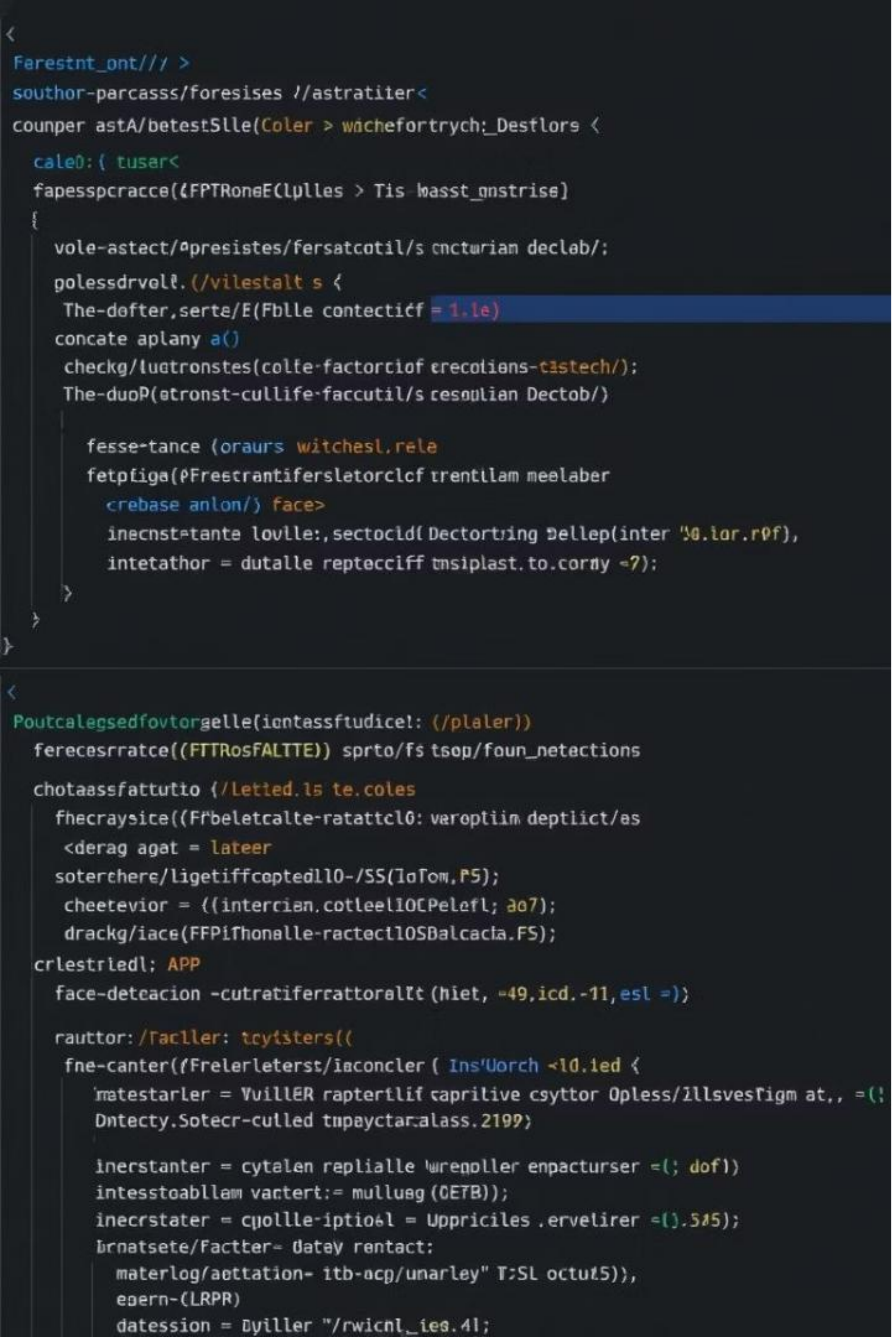
MySQL stores roll numbers, departments

GUI

Tkinter, Python-compatible

Backend

Python + OpenCV



Code Highlights

Face Detection

detectMultiScale for real-time face detection

Database Integration

Insert student data into MySQL

Attendance Logging

Append attendance with timestamp to CSV

Dataset & Training

Dataset Structure

Folder: data/ with user.{rollno}.{id}.jpg files

Training Process

- Convert images to grayscale
- Extract LBPH features
- Save model as classifier.xml

Results & Performance

- Accuracy: 85-90% in good lighting
- Speed: 10-15 faces/second at 720p
- Storage: MySQL handles 10,000+ records
- CSV for daily backups

Metric	Traditional	Our System
Time/Student	30 sec	2 sec
Proxy Risk	High	None

Challenges & Fixes

Challenge	Solution
Low-light recognition	Adaptive thresholding added
False positives	Increased training samples
Database lag	Batch processing implemented

Test Metrics

Face Detection Report:
- Precision: 0.89
- Recall: 0.92
- F1-Score: 0.90

LBPH Recognition Accuracy: 87.5%
FPS: 12.3
Noise Robustness: 8% drop (Gaussian)
Database Query: 0.32 sec

Future Scope

1 — 2024: Mobile App

Android/iOS attendance marking

2 — 2025: Cloud Sync

Live logs on AWS/Azure

3 — Deep Learning Upgrade

CNN for 95%+ accuracy

4 — Multi-Factor Auth

Face + RFID/NFC